DESCRIPTION OF COURSEWORK

Course Code	EEE308
Course Name	RF&Microwave Circuits
Lecturer	Associate Prof. Dr. Zulkifly Abbas
Academic Session	2025/09
Assessment Title	Assignment : Design a distributed branch line coupler at 2 GHz and simulate the performance using a RF/Microwave circuit design software.

A. Introduction/ Situation/ Background Information

A coupler is a device that couples the power from the input port to two or more output ports equally with less loss and with or without the phase difference. A branch line coupler is a 3 dB coupler with a 90º-phase difference between the two output ports. An ideal branch line coupler, as shown in Figure 1, is a four-port network and is perfectly matched at all four ports.

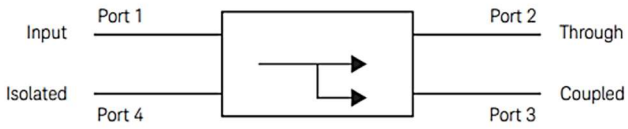


Figure 1. Ideal coupler

The power entering port 1 is evenly divided between ports 2 and 3, with a phase shift of 90 degree between the ports. The 4th port is the isolated port and no power flows through it. The branch line coupler has a high degree of symmetry and allows any of the four ports to be used as the input port. The output ports are on the opposite side of the input port and the isolated port is on the same side of the input port. This symmetry is reflected in the S matrix, as each row can be the transposition of the first row. The [S] matrix of the ideal branch line coupler is given as follows:

$$[S]=\frac{-1}{\sqrt{2}}\begin{bmatrix}0 & j & 1 & 0\\ j & 0 & 0 & 1\\ 1 & 0 & 0 & j\\ 0 & 1 & j & 0\end{bmatrix}$$

The major advantage of this coupler is easier realization and the disadvantages are lesser bandwidth due to the use of a quarter wavelength transmission line for realization and

discontinuities occurring at the junction. To circumvent the above disadvantages, cascading multiple sections of the branch line coupler can increase the bandwidth by a decade and a 10º – 20º increase in length of the shunt arm can compensate for the power loss due to discontinuity effects.

Objective:

Design a distributed branch line coupler at 2 GHz and simulate the performance using a RF/Microwave circuit design software.

Methodology

The basic steps to create a distributed model for the branch line coupler is as follows:

1. Select appropriate values for the design of the coupler. For this assignment, we will select following dielectric parameters: -

   $\varepsilon_r = 4.2$

   Height = 1.5 mm

   Loss Tangent = 0.0019

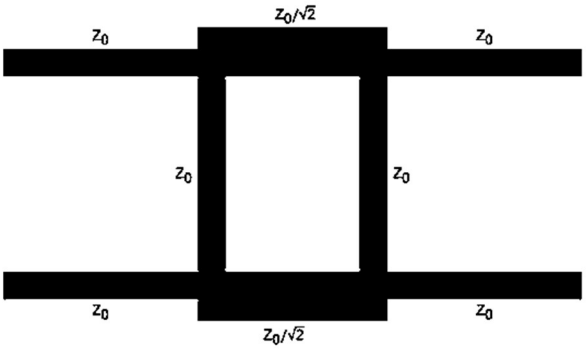
   Metal Thickness = 0.035 mm

   Metal Conductivity = 5.8E7 S/m

2. Calculate the wavelength λg from the given frequency specifications as follows:

   $\lambda_g = \frac{c}{f\sqrt{\epsilon_r}}$

3. Synthesize the physical parameters (length and width) for the λ/4 lines with impedances of Z0 and Z0/√2 (Z0 is the characteristic impedance of the microstrip line which is taken as 50 Ω). The geometry of the Branch line coupler is shown below



. Geometry of branch line coupler

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Build(design) microwave circuits and measure(lab) its signals

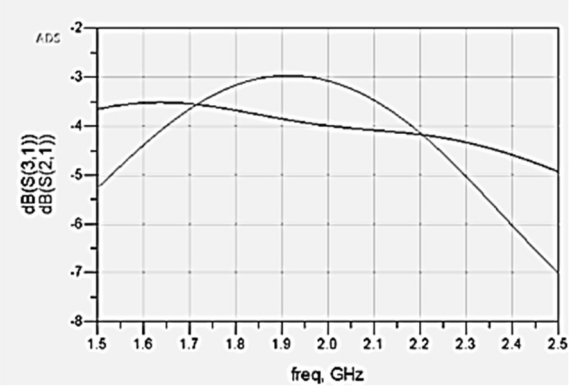
C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

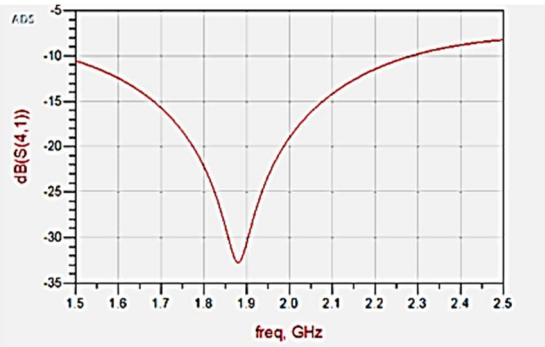
2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

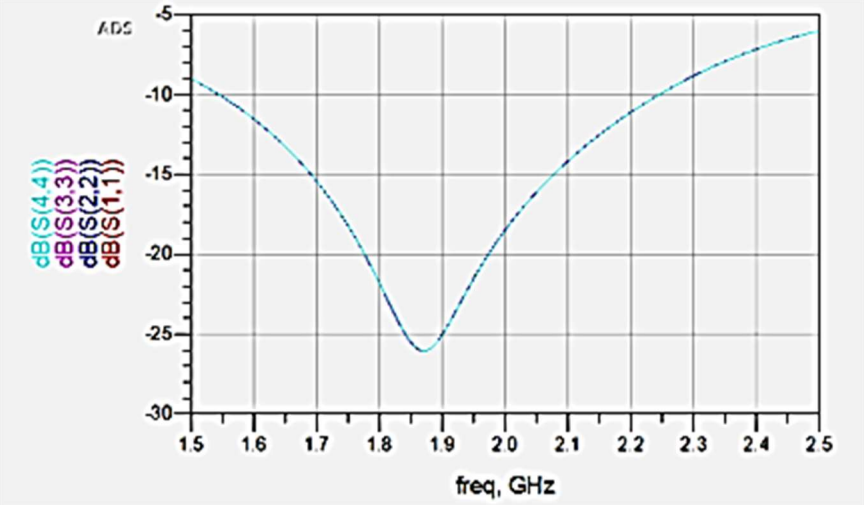
Setup the simulation frequency from 1.5 GHz to 2.5 GHz. Once the simulation is finished, plot the output parameters – dB(S(2,1)) and dB(S(3,1)) – on one graph; the isolation port – dB(S(2,1)) – on another graph; and the reflection parameters – dB(S(1,1)), dB(S(2,2)), dB(S(3,3)), and dB(S(4,4)) – on another graph. Examples of expected results are shown below



Output parameters



Isolation port



Reflection parameters

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Design a distributed branch line coupler at 2 GHz and simulate the performance using a RF/Microwave circuit design software.	100 %
	TOTAL	**100 %**

F. Tasks

Design a distributed branch line coupler at 2 GHz and simulate the performance using a RF/Microwave circuit design software.

The report must be submitted (softcopy/ hardcopy) before June 13, 2025. The assignment report must have the components outlined in the rubrics in Appendix 1 and submitted before end of week 14.

APPENDIX 1

MARKING RUBRICS

	Excellent	Good	Adequate	Poor	Lacking
Abstract/ Introduction/ Theory	1.Includes complete well-written information on the background to understand the subject of the experiment and its significance. 2.Summarizes what is already known on the subject. 3. Includes the question to be answered by the lab and states hypothesis that is based on research and/or sound reasoning, described in the background. 4. Includes the description on how the hypothesis is going to be tested. 5. Deep understanding of the subject as well as creativity is demonstrated.	1.Includes strong introductory information, most major points provided. 2.Most important details in the summary provided. 3. Most important details of the hypothesis covered. 4. An idea on how to test the hypothesis is included. 5. Good understanding is demonstrated.	1.Includes good introductory information, missing some points in the summary. 2.Hypothesis is missing some minor details. 3.No description on how the hypothesis is going to be tested is provided. 4. Appropriate understanding of the subject is demonstrated.	1. Introductory information provided, missing some major points. 2. Important details of the hypothesis are missing. 3. No description on how the hypothesis is going to be tested provided. 4. Limited understanding of the subject is demonstrated.	1. Part is missing or plagiarized. OR 2.Little background information is present. 3.Incorrect information is provided. 4. For the hypothesis, important experimental details are missing, of the hypothesis is missing. 5. Poor understanding or no understanding of the subject is demonstrated.
methodology	Description (not a copy of a protocol!) is included, detailed enough so it could be repeated by another scientist. Description is not overloaded with unnecessary details.	Description included, some steps are vague or unclear OR instead of description, a step-by–step protocol is copied from manual.	The description gives generalities, enough for reader to understand how the experiment was conducted	Would be impossible to repeat the experiment, reader must guess at how the data was gathered or experiment conducted	Part is missing or plagiarized

Results	Results and data are clearly recorded, so it is easy for the reader to see trends. Excellent graphs, tables or figures with appropriate names and labels, excellent information and understanding of results, excellent interpretation of data and trends while referring to figures and tables.	Results are clear, but trends are not obvious or there are minor errors in organization. Good figures, graphs, or tables, good information results, good understanding of results, but some figure/table names or labels are missing.	Results are unclear, trends are not obvious, disorganized, but there is enough data to show the experiment was conducted. Appropriate figures, graphs or tables or drawings, some labels and figure names are missing. Only some understanding of results is evident.	Results are disorganized or poorly recorded, do not make sense; not enough data was taken to justify results. Missing or incorrect information.	Part is missing or plagiarized. Misleading, confusing and/or incorrect information. Poorly presented results.
Discussion/ Conclusions	1.Summarizes data used to draw conclusions 2.Conclusions follow data (not wild guesses or leaps of logic) 3. Hypothesis is rejected or accepted based on the data. 4. Discussion of applications or realworld connections is present. 5. Deep understanding of the subject as well as creativity is demonstrated.	Good conclusion, important information provided, or one point from Excellent is missing	Appropriate conclusion. Two points from Excellent are missing.	Limited conclusion. Important information is missing.	Part is missing or plagiarized. OR No conclusion is made. Missing important information. Little effort or reflection is shown.
References Formatting	All references and citations are correctly written and present. More than two resources are cited. Student accurately cites to the sources throughout the text.	One reference or citation is missing or incorrectly present.	Appropriate citing and reference list.	References and citations are missing or errors detract significantly from paper.	Part is missing or plagiarized. Incorrect citing.

Component Title	ASSIGNMENT				Percentage (%)	100
Criteria	Score and Descriptors					Weight (%) Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)	
Abstract						10
Introduction & Theory						10
Methodology/ Experimental						35
Results, Disscussion Conclusion						40
Formatting References						5
					TOTAL	100